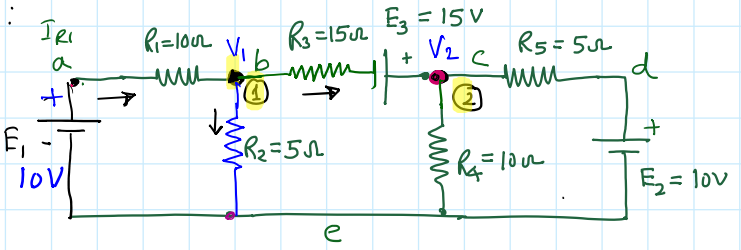
Example 5:

Now let us consider the following circuit:

Find the currents in R_1, R_2, R_3, R_4, R_5 using node-pair method.



$J = 3$

n = No of unknown variables $= J-1 = 2$

KCL @ Node 1: $\dfrac{V_1 - E_1}{R_1} + \dfrac{V_1 - 0}{R_2} + \dfrac{V_1 - (-E_3 + V_2)}{R_3} = 0$

$$\Rightarrow \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) V_1 - \frac{V_2}{R_3} = \frac{E_1}{R_1} + \frac{(-E_3)}{R_3} \quad \text{---- (1)}$$

KCL at node 2:

$$\frac{V_2 - (V_1 + E_3)}{R_3} + \frac{V_2}{R_4} + \frac{V_2 - E_2}{R_5} = 0$$

$$-\frac{1}{R_3} V_1 + \left(\frac{1}{R_3} + \frac{1}{R_4} + \frac{1}{R_5}\right) V_2 = \frac{E_2}{R_5} + \frac{E_3}{R_3} \quad \text{--- (2)}$$

Equations (1) and (2) can be solved algebrically.

We can also express in matrix format, as given below.

$$[G][V] = [I]$$

$$\begin{bmatrix} G_{11} & G_{12} \cdots & G_{1n} \\ G_{21} & G_{22} \cdots & G_{2n} \\ \vdots & & \\ G_{n1} & G_{n2} \cdots & G_{nn} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_n \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_n \end{bmatrix}$$

where $G_{ii} = \sum_{j=1}^{n} g_{ij}$ and $G_{ij} = -g_{ij}$

$= $ Self conductance at node i

and $G_{ij} = -g_{ij} =$ Mutual conductance between junctions i & j.

For this example

$$\begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} E_1/R_1 - E_3/R_3 \\ E_2/R_5 + E_3/R_3 \end{bmatrix}$$

$$G_{11} = g_{11} + g_{12} + g_{13} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{10} + \frac{1}{5} + \frac{1}{15} = 11/30$$

$\dfrac{3+6+2}{30}$

$0.1 + 0.2 +$

$G_{11} = g_{11} + g_{12} + g_{13} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{10} + \frac{1}{5} + \frac{1}{15} = 11/30$

$G_{12} = G_{21} = -g_{12} = -g_{21} = -\frac{1}{R_3} = -\frac{1}{15}$

$G_{22} = g_{21} + g_{22} + g_{23} = \frac{1}{R_3} + \frac{1}{R_4} + \frac{1}{R_5} = \frac{1}{15} + \frac{1}{10} + \frac{1}{5} = 0.36\ U$

$I_1 = \frac{E_1}{R_1} - \frac{E_3}{R_3} = \frac{10}{10} - \frac{15}{15} = 0$

$I_2 = \frac{E_3}{R_3} + \frac{E_2}{R_5} = \frac{15}{15} + \frac{10}{5} = 3$

$$\Rightarrow \begin{bmatrix} 11/30 & -1/15 \\ -1/15 & 11/30 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 11/30 & -1/15 \\ -1/15 & 11/30 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 1.538 \\ 8.462 \end{bmatrix}$$

$\Rightarrow$ $V_1 = 1.538\ V;\quad V_2 = 8.462\ V$

Once V_1 and V_2 are known, we can find currents and other electrical parameters such as power.

$I_1 = \frac{E_1 - V_1}{R_1} = \frac{10 - 1.538}{10} = 0.846\ A$

$I_{R2} = \frac{V_1}{R_2} = \frac{1.538}{5} = 0.307\ A$

$I_{R3} = \frac{V_1 - (-E_3 + V_2)}{R_3} \quad \frac{1.538 - (-15 + 8.462)}{15} = 0.538\ A$

$I_{R4} = \frac{V_2}{R_4} = \frac{8.462}{10} = 0.8462\ A$

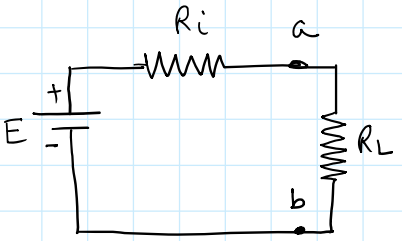
$I_{R5} = \frac{V_2 - E_2}{R_5} = \frac{8.462 - 10}{5} = -0.308\ A$

The solution is the same as we obtained by the mesh method, i.e. loopwise $I_1 = I_{R1} = 0.846A$, $I_2 = I_{R2} = 0.538\ A$, $I_3 = I_{R5} = -0.308\ A$

Thevenin's Theorem and Norton Theorem:

Consider the shown circuit.

Consider the shown circuit.



Two important parameters are:

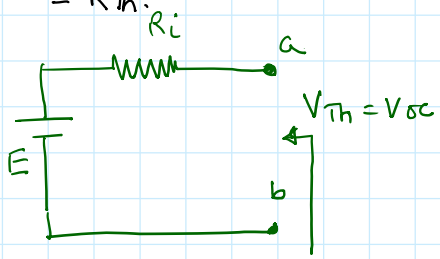
1) Open circuit voltage (Voc)
2) short circuit current (Isc)

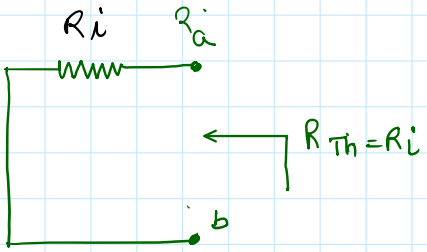
Voc = Oper circuit voltage = Thevenin's voltage = V_{Th}

R_i = Resistance looking terminal ab with all sources inactive.

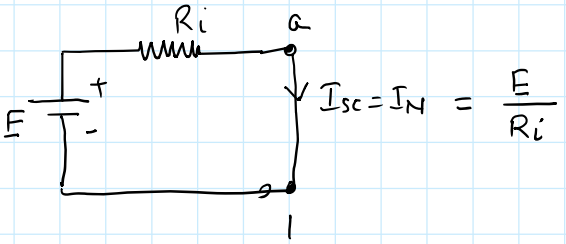
= Thevenin's resistance

= R_{Th}.





Isc = short circuit current = Norton Current = I_N



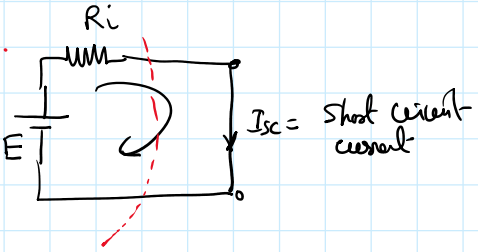
The the relation between $V_{oc} = V_{Th}$ and $I_{sc} = I_N$ is: $\frac{V_{oc}}{I_{sc}} = \frac{V_{Th}}{I_N} = R_i$

Inactive source → voltage source → Short them

Current source → Open them.

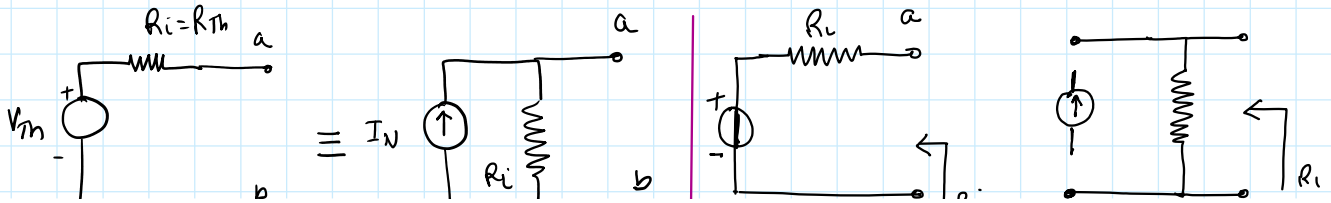
$$I_{sc} = \frac{E}{R_i} = \frac{V_{oc}}{R_i} = \frac{V_{Th}}{R_{Th}}$$

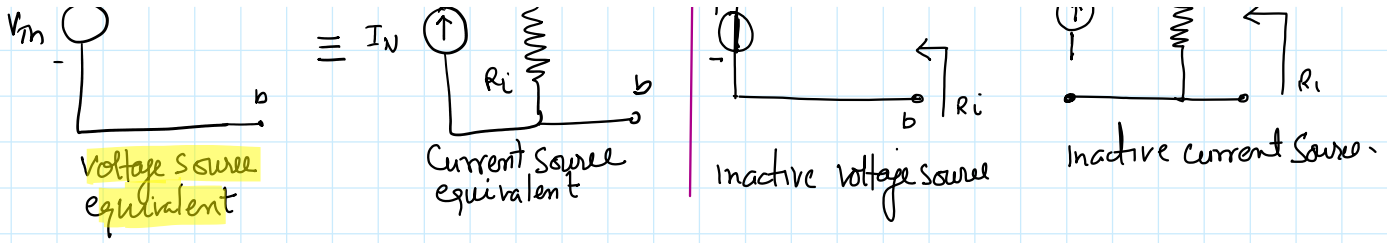
$$I_{Norton} = \boxed{I_N = \frac{V_{Th}}{R_{Th}}}$$



Norton's Theorem

$I_N = I_{sh}$





Voltage source equivalent

Current Source equivalent

Inactive voltage source

Inactive current Source.

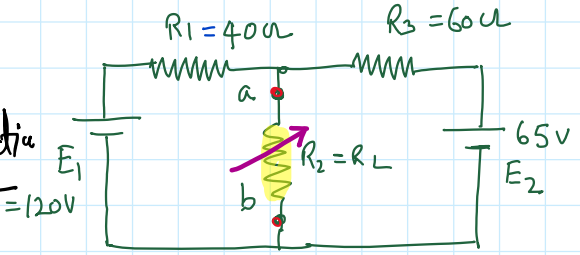
$V_{Th} = I_N * R_i$

$V_{OC} = I_{SC} * R_i$

$$R_i = \frac{V_{OC}}{I_{SC}}$$

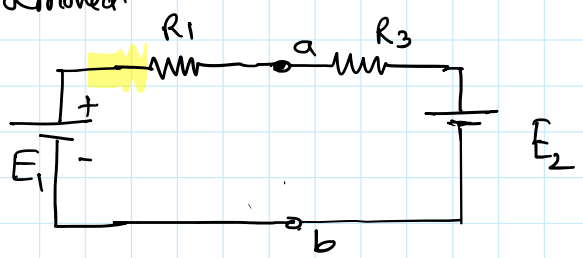
$$I_N = \frac{V_{Th}}{R_L} = \frac{V_{OC}}{R_i}$$

Example 6: Find current in resistance R_2 using Thevenin's and Norton Equivalents.



Thevenin Equivalent Ckt:
Open circuit voltage Representation

$V_{OC} = V_{Th} = V_{ab}$ | when Load is removed



$$I = \frac{E_1 - E_2}{R_1 + R_3} = \frac{120 - 65}{40 + 60}$$

$$I = 0.55\ A$$

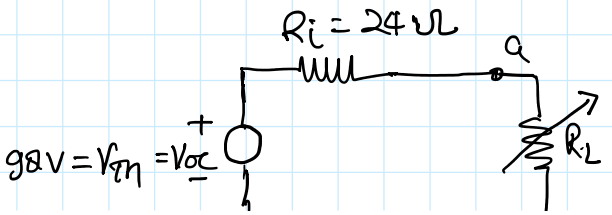
$V_{ab} = 120 - 0.55 \times 40 = 98\ V$

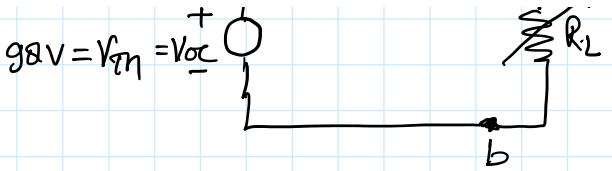
$= R_3 \times I + E_2 = 60 \times 0.55 + 65 = 98\ V$

$R_i = R_{Th}$ = Resistance looking into terminal ab with load removed with all sources inactive.

$$R_i = R_{Th} = \frac{R_1 R_3}{R_1 + R_3} = \frac{40 \times 60}{40 + 60} = 24\ \Omega$$

Thus the Thevenin equivalent circuit is





Thevenin's equivalent circuit

Now the current through $R_2 = R_L$ is given by:

$$I_{R_2} = I_{R_L} = \frac{V_{Th}}{R_{Th}+R_L} = \frac{98}{24+R_L} = \frac{98}{24+20} = 2.23\,A$$

which is same as previously computed, using mesh or nodal analysis.

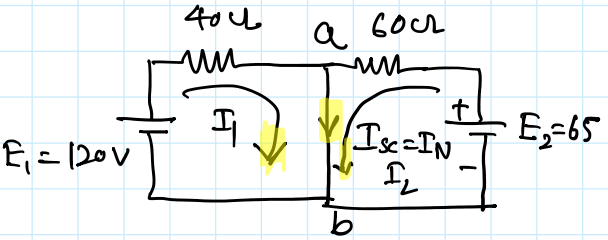
Norton Equivalent Circuit:

What is short **circuit** when the load is shorted?

$I_{sc} = I_N = ?$

$I_1 = \frac{120}{40} = 3\,A$

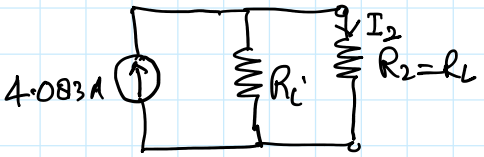
$I_2 = \frac{65}{60} = 1.083$



$$I_{sc} = I_1 + I_L = 4.083 = \frac{V_{oc}}{R_i} = \frac{98}{24} = 4.083\,A$$

$R_i = 24\Omega$

$$I_{R_2} = (I) * \frac{R_i}{R_i+R_L}$$



$$= \frac{4.083 \times 24}{24+20} = 2.2\,A \text{ (same as earlier)}$$

Thevenin's Theorem:

Any two terminal network composed of linear passive and active elements, can be replaced by an equivalent voltage source and an equivalent series resistance. The voltage source is equal to the voltage across the two terminals, caused by active elements, with load element removed (open). The series resistance is the equivalent resistance looking into the two terminals with all power sources inactive.

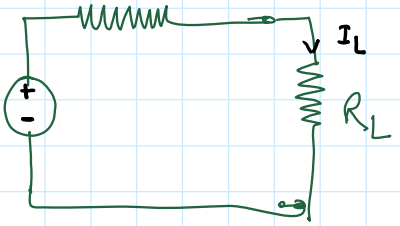
$R_{Th} = R_i$

equivalent resistance looking into the two terminals with all power sources inactive.

$R_{Th} = R_i$

$V_{OC} = V_{Th}$
$R_i = R_{Th}$

$V_{OC} = V_{Th}$



Norton's Theorem : Dual of Thevenin's equivalent circuit.

Any two terminal network composed of linear passive and active elements, can be replaced by an equivalent current source and an equivalent shunt resistance. The current source is equal to the current into the two terminals, caused by active elements, with load element shorted.

The shunt resistance is the equivalent resistance looking into the two terminals with all power sources inactive.